clk
rst_n
rd_en

wr_en
data_in[31:0]

!=wr_ptr2_i RTL_NEQ O

I0[4:0]
I1[4:0]

wr_ptr2_i__0 RTL_AND

I0

I1 O

rd_ptr0_i RTL_AND

I0

I1 O

wr_ptr1_i RTL_AND

I0

I1 O

count_i__0
RTL_MUX

S=1'b1 I0

S=default I1 O S+

count0_i RTL_ADD

I1

I0[4:0] O[4:0] -

count0_i__0 RTL_SUB

I1

I0[4:0] O[4:0]

count_i__1
RTL_MUX

S=1'b1 I0

S=default I1 O S

count_i
RTL_MUX S

S=1'b1 I0[4:0] S=default I1[4:0] O[4:0]

count_reg[4:0]

RTL_REG_ASYNC C

CE

CLR D Q

!=wr_ptr4_i RTL_NEQ O

I0[4:0] V=B"10000" I1[4:0]

wr_ptr3_i RTL_AND

I0

I1 O

wr_ptr_i
RTL_MUX

S=1'b1 I0

S=default I1 O S+wr_ptr0_i RTL_ADD

I1

I0[3:0] O[3:0]

mem_i
RTL_MUX

S=1'b1 I0

S=default I1 O S

wr_ptr_reg[3:0]

RTL_REG_ASYNC C

CE

CLR D Q

RTL_ROM__0 RTL_ROM

A[3:0] O

RTL_ROM__1 RTL_ROM

A[3:0] O

RTL_ROM__2 RTL_ROM

A[3:0] O

RTL_ROM__3 RTL_ROM

A[3:0] O

RTL_ROM__4 RTL_ROM

A[3:0] O

RTL_ROM__5 RTL_ROM

A[3:0] O

RTL_ROM__6 RTL_ROM

A[3:0] O

RTL_ROM__7 RTL_ROM

A[3:0] O

RTL_ROM__8 RTL_ROM

A[3:0] O

RTL_ROM__9 RTL_ROM

A[3:0] O

RTL_ROM__10 RTL_ROM

A[3:0] O

RTL_ROM__11 RTL_ROM

A[3:0] O

RTL_ROM__12 RTL_ROM

A[3:0] O

RTL_ROM__13 RTL_ROM

A[3:0] O

RTL_ROM__14 RTL_ROM

A[3:0] O

mem_i__0
RTL_MUX

S=1'b0 I0

S=default I1 O S

RTL_ROM
RTL_ROM

A[3:0] O
rd_ptr_i
RTL_MUX

S=1'b1 I0

S=default I1 O S

RTL_AND__0 RTL_AND

I0

I1 O RTL_AND__1 RTL_AND

I0

I1 O RTL_AND__2 RTL_AND

I0

I1 O RTL_AND__3 RTL_AND

I0

I1 O RTL_AND__4 RTL_AND

I0

I1 O RTL_AND__5 RTL_AND

I0

I1 O RTL_AND__6 RTL_AND

I0

I1 O RTL_AND__7 RTL_AND

I0

I1 O RTL_AND__8 RTL_AND

I0

I1 O RTL_AND__9 RTL_AND

I0

I1 O RTL_AND__10 RTL_AND

I0

I1 O RTL_AND__11 RTL_AND

I0

I1 O RTL_AND__12 RTL_AND

I0

I1 O RTL_AND__13 RTL_AND

I0

I1 O RTL_AND__14 RTL_AND

I0

I1 O

RTL_MUX__0
RTL_MUX

S=4'b1110 I0[31:0] O[31:0] S[3:0] RTL_MUX__1
RTL_MUX

S=4'b1101 I0[31:0] O[31:0] S[3:0] RTL_MUX__2
RTL_MUX

S=4'b1100 I0[31:0] O[31:0] S[3:0] RTL_MUX__3
RTL_MUX

S=4'b1011 I0[31:0] O[31:0] S[3:0] RTL_MUX__4
RTL_MUX

S=4'b1010 I0[31:0] O[31:0] S[3:0] RTL_MUX__5
RTL_MUX

S=4'b1001 I0[31:0] O[31:0] S[3:0] RTL_MUX__6
RTL_MUX

S=4'b1000 I0[31:0] O[31:0] S[3:0] RTL_MUX__7
RTL_MUX

S=4'b0111 I0[31:0] O[31:0] S[3:0] RTL_MUX__8
RTL_MUX

S=4'b0110 I0[31:0] O[31:0] S[3:0] RTL_MUX__9
RTL_MUX

S=4'b0101 I0[31:0] O[31:0] S[3:0] RTL_MUX__10
RTL_MUX

S=4'b0100 I0[31:0] O[31:0] S[3:0] RTL_MUX__11
RTL_MUX

S=4'b0011 I0[31:0] O[31:0] S[3:0] RTL_MUX__12
RTL_MUX

S=4'b0010 I0[31:0] O[31:0] S[3:0]

rd_ptr_i__0
RTL_MUX

S=1'b1 I0

S=default I1 O S+

rd_ptr0_i__0 RTL_ADD

I1

I0[3:0] O[3:0]

RTL_AND
RTL_AND

I0

I1 O

data_out_i
RTL_MUX

S=1'b1 I0

S=default I1 O S

RTL_MUX
RTL_MUX

S=4'b1111 I0[31:0] O[31:0] S[3:0] RTL_MUX__13
RTL_MUX

S=4'b0001 I0[31:0] O[31:0] S[3:0] RTL_MUX__14
RTL_MUX

S=4'b0000 I0[31:0] O[31:0] S[3:0]

mem_reg[0][31:0]

RTL_REG_ASYNC C

CE

CLR

n/c D

PRE

n/c Q

mem_reg[1][31:0]

RTL_REG_ASYNC C

CE

CLR

n/c D

PRE

n/c Q

mem_reg[2][31:0]

RTL_REG_ASYNC C

CE

CLR

n/c D

PRE

n/c Q

mem_reg[3][31:0]

RTL_REG_ASYNC C

CE

CLR

n/c D

PRE

n/c Q

mem_reg[4][31:0]

RTL_REG_ASYNC C

CE

CLR

n/c D

PRE

n/c Q

mem_reg[5][31:0]

RTL_REG_ASYNC C

CE

CLR

n/c D

PRE

n/c Q

mem_reg[6][31:0]

RTL_REG_ASYNC C

CE

CLR

n/c D

PRE

n/c Q

mem_reg[7][31:0]

RTL_REG_ASYNC C

CE

CLR

n/c D

PRE

n/c Q

mem_reg[8][31:0]

RTL_REG_ASYNC C

CE

CLR

n/c D

PRE

n/c Q

mem_reg[9][31:0]

RTL_REG_ASYNC C

CE

CLR

n/c D

PRE

n/c Q

mem_reg[10][31:0]

RTL_REG_ASYNC C

CE

CLR

n/c D

PRE

n/c Q

mem_reg[11][31:0]

RTL_REG_ASYNC C

CE

CLR

n/c D

PRE

n/c Q

mem_reg[12][31:0]

RTL_REG_ASYNC C

CE

CLR

n/c D

PRE

n/c Q

mem_reg[13][31:0]

RTL_REG_ASYNC C

CE

CLR

n/c D

PRE

n/c Q

mem_reg[14][31:0]

RTL_REG_ASYNC C

CE

CLR

n/c D

PRE

n/c Q

rd_ptr_reg[3:0]

RTL_REG_ASYNC C

CE

CLR D Q

data_out_i__0
RTL_MUX

S=1'b1 I0

S=default I1 O S

mem_reg[15][31:0]

RTL_REG_ASYNC C

CE

CLR

n/c D

PRE

n/c Q

RTL_MUX__15

RTL_MUX

S=4'b1111 I0[31:0] S=4'b1110 I1[31:0] S=4'b0101 I10[31:0] S=4'b0100 I11[31:0] S=4'b0011 I12[31:0] S=4'b0010 I13[31:0] S=4'b0001 I14[31:0] S=4'b0000 I15[31:0] S=4'b1101 I2[31:0] S=4'b1100 I3[31:0] S=4'b1011 I4[31:0] S=4'b1010 I5[31:0] S=4'b1001 I6[31:0] S=4'b1000 I7[31:0] S=4'b0111 I8[31:0] S=4'b0110 I9[31:0]

O[31:0]

S[3:0]

data_out_i__1
RTL_MUX

S=1'b0 I0

S=default I1 O S

<=

almost_empty0_i

RTL_LEQ O

I0[4:0] V=B"00001" I1[4:0]

>=

almost_full0_i

RTL_GEQ O

I0[4:0] V=B"01110" I1[4:0] =

empty0_i RTL_EQO

I0[4:0]

I1[4:0] =

full0_i RTL_EQO

I0[4:0] V=B"10000" I1[4:0]

data_out_reg[31:0]

RTL_REG_ASYNC C

CE

CLR

n/c D

PRE

n/c Q

almost_empty_reg

RTL_REG_ASYNC C

CE

CLR

n/c D

PRE

n/c Q
almost_full_reg

RTL_REG_ASYNC C

CE

CLR

n/c D

PRE

n/c Q
empty_reg

RTL_REG_ASYNC C

CE

CLR

n/c D

PRE

n/c Q
full_reg

RTL_REG_ASYNC C

CE

CLR

n/c D

PRE

n/c Q

data_out[31:0]

almost_empty

almost_full

empty

full